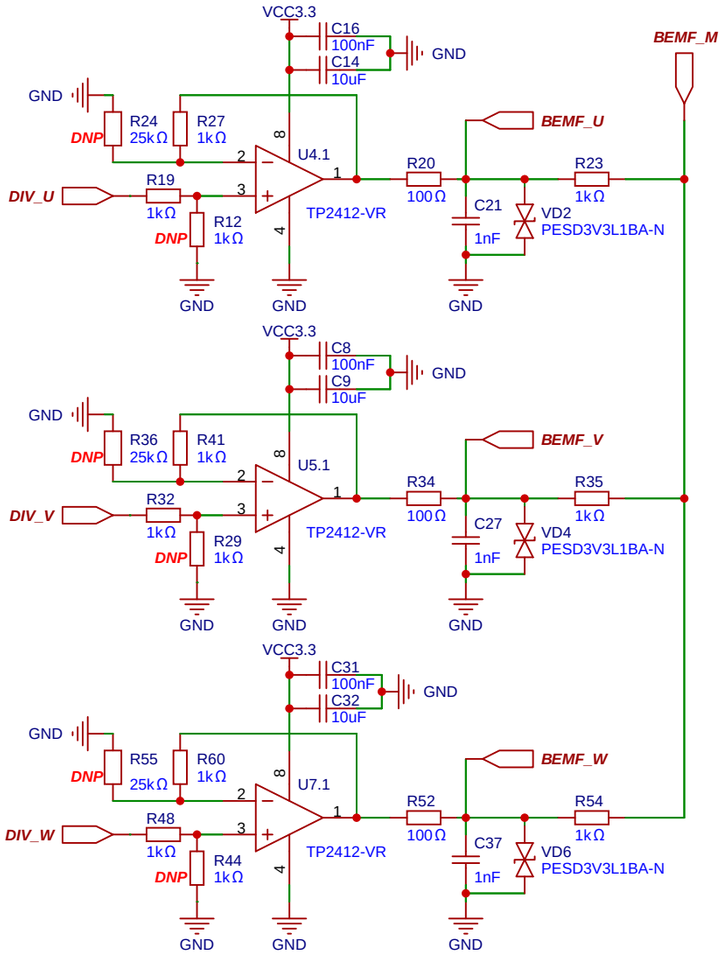
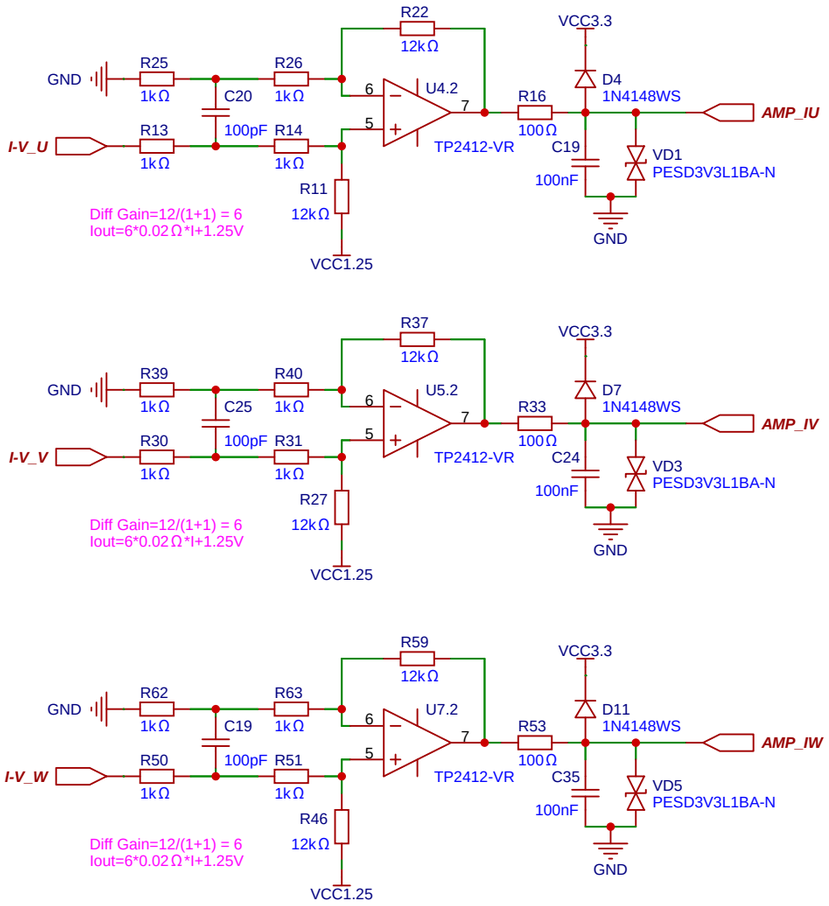
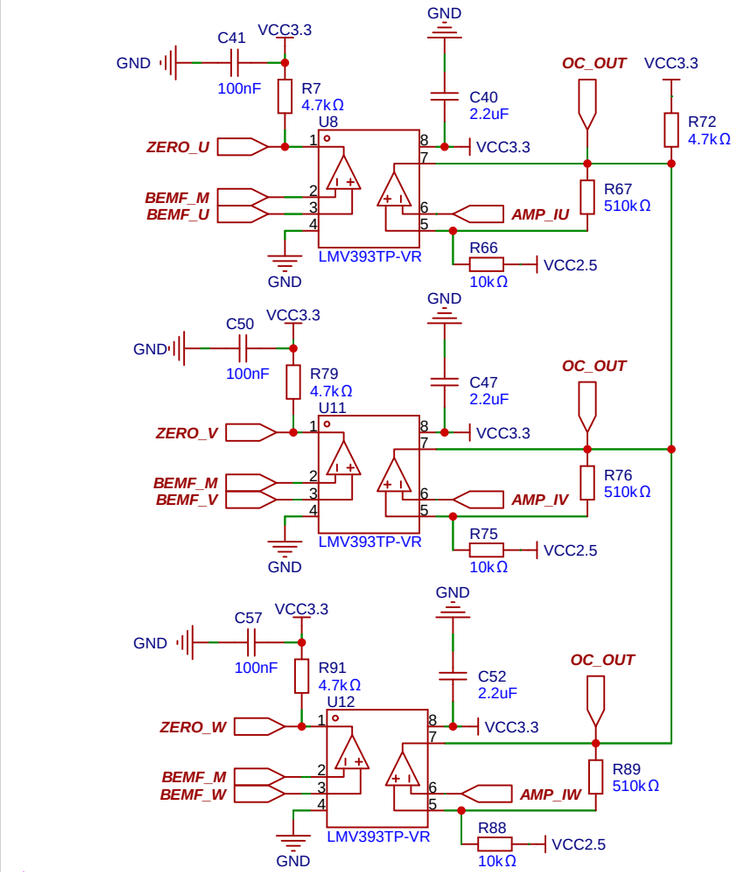
电压&电流采集



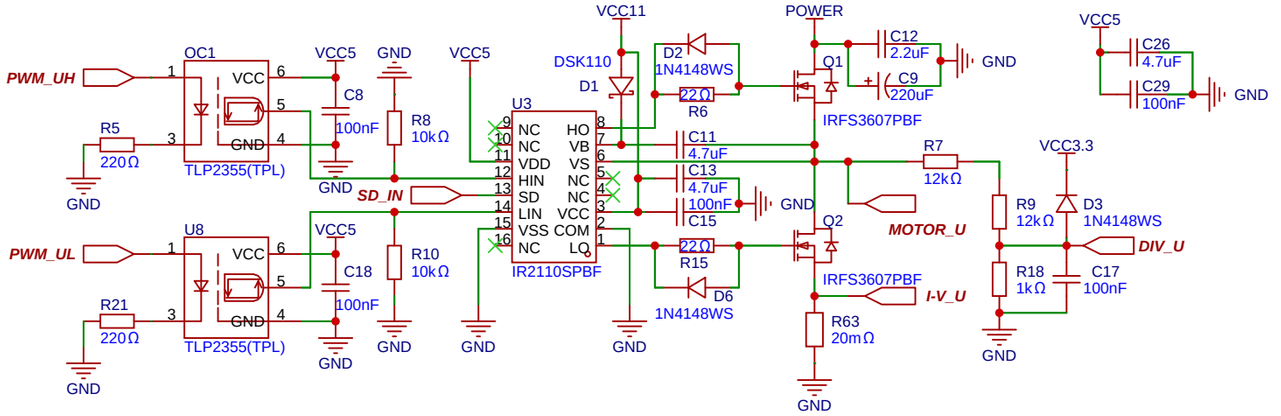


过零&过流检测

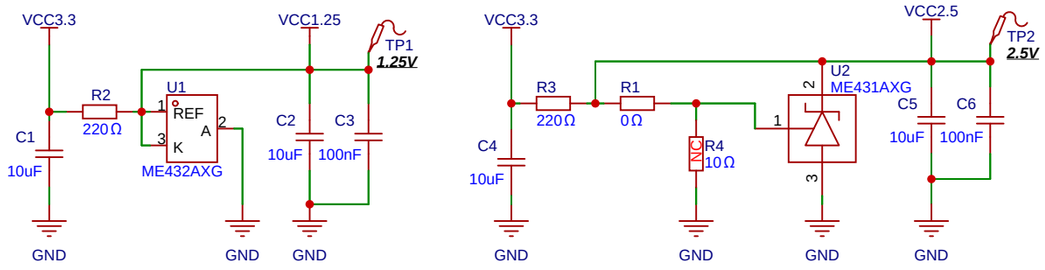


△U= 10K/520K * 3.3V=64mV
△A= 64mV/(2.5V-1.25V) *10A= 0.51A
△U= R1/(R1+R2)*VDD
R1=10K;R2=510K;VDD=3.3V

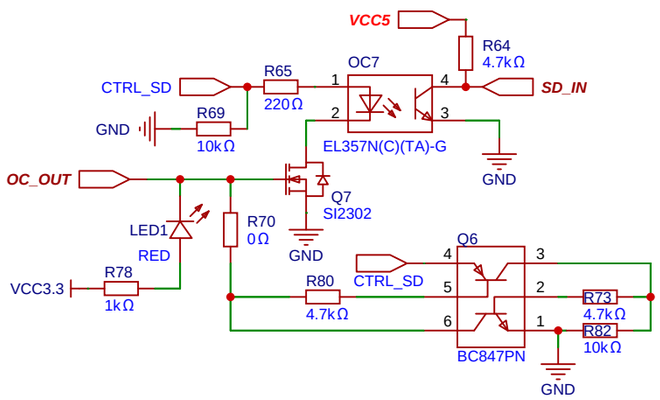
三相H桥



参考电压



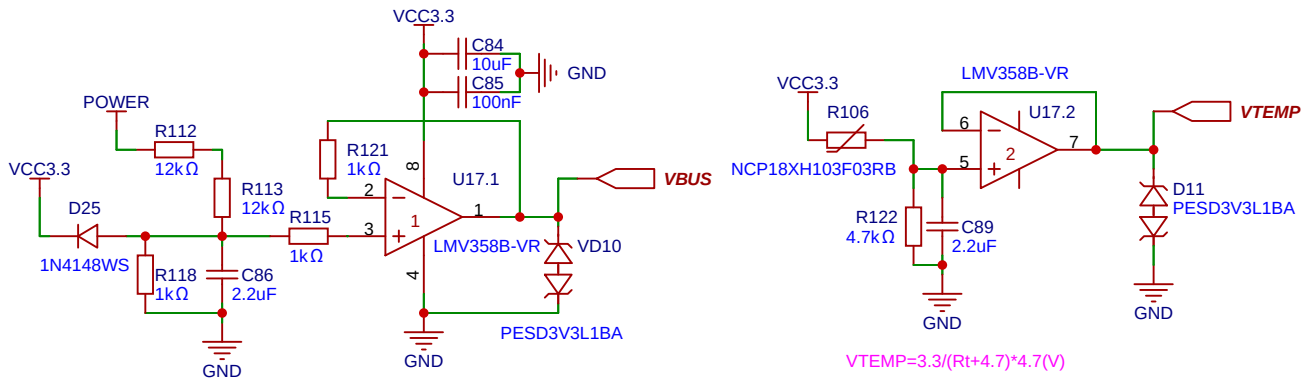
关断&过流保护



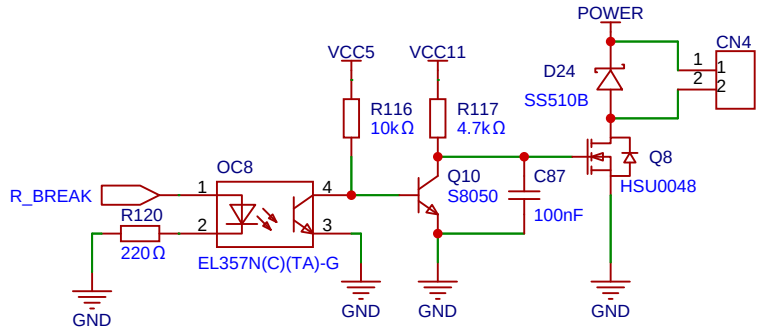
焊接R72,带锁存保护: CTRL_SD=0, H桥无效; CTRL_SD=1:
未发生过流(OC_OUT=1)时,光耦导通,H桥有效; 当发生过流
(OC_OUT=0)时,在自锁电路作用下OC_OUT恒为低(即使解除
过流状态),光耦关断,H桥关断,直到CTRL_SD=0时，清除锁存状态；
不焊接R70,周期保护: CTRL_SD=0, H桥无效; CTRL_SD=1,
过流时H桥无效，解除过流状态时H桥有效;

原理图	Schematic1	更新日期	2025-07-15
		创建日期	2025-01-23
图页	P1	物料编码	
绘制		Motor-1_配套无刷电机驱动板	
审阅			
		版本	尺寸
		V1.0	
嘉立创EDA		Genbotter	

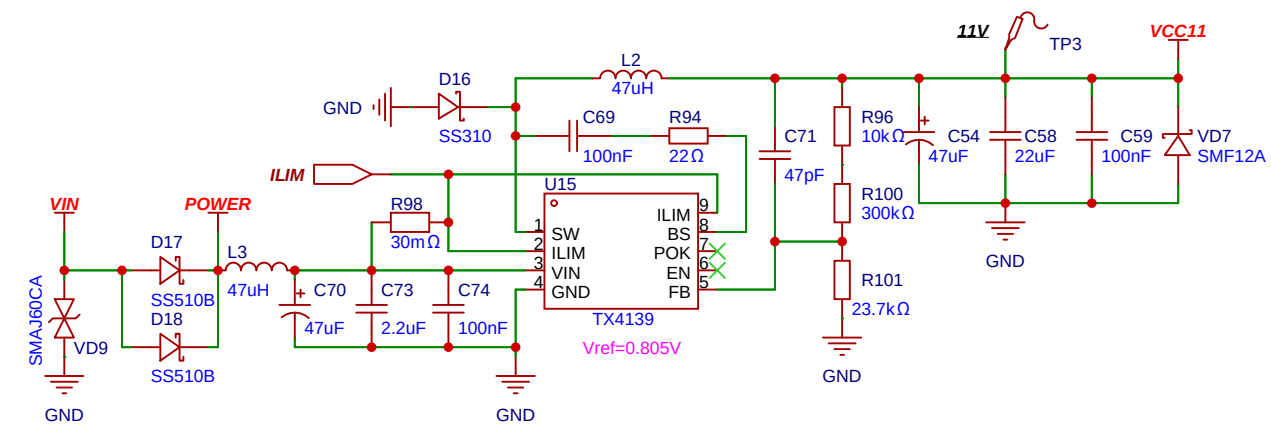
电源&温度采集

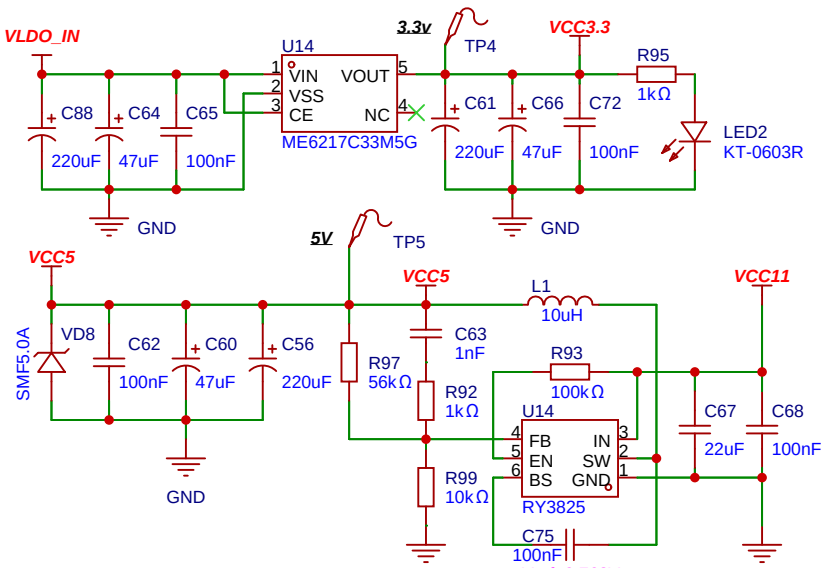


制动电路

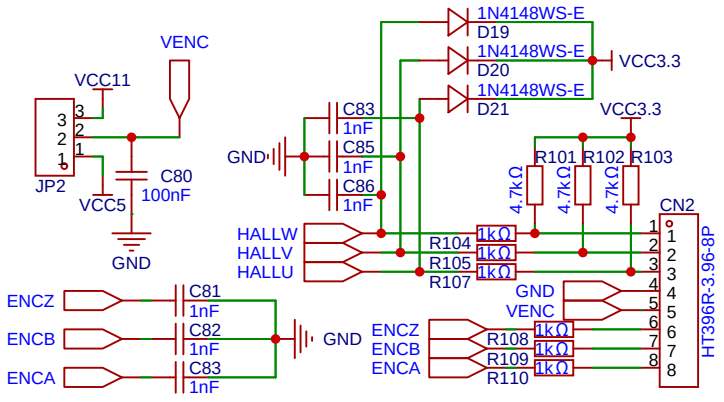


电源系统

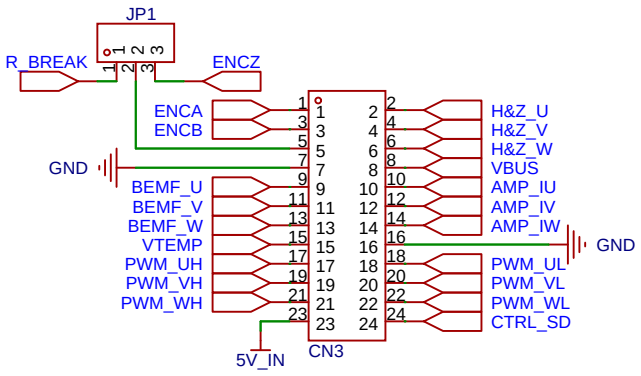




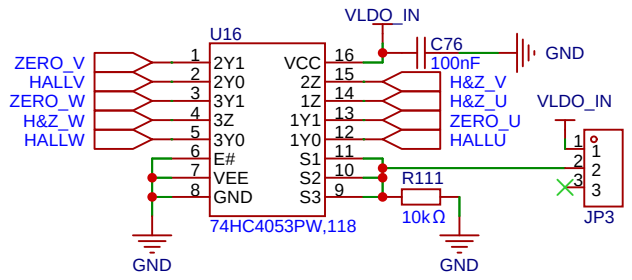
霍尔&编码器接口



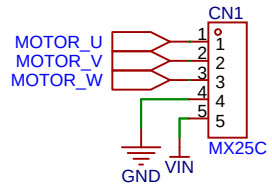
控制&采集接口



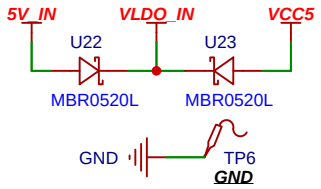
霍尔&过零检测复用



接线端子



LDO供电电源



原理图	Schematic1	更新日期	2025-07-15
		创建日期	2025-01-24
图页	P2	物料编码	
绘制		Motor-1_配套无刷电机驱动板	
审阅			
		版本	尺寸	页 2 共 2
嘉立创EDA		V1.0	A4	嘉立创EDA